

Ashwini Kumar Sharma

Senior Data Engineer

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Professional Summary

Senior Data Engineer with 15+ years of experience designing and delivering scalable, production-ready data pipelines, ELT/ETL frameworks, and cloud data warehouse architectures. Deep hands-on expertise with Snowflake, Databricks, BigQuery, Redshift, dbt (Data Build Tool), Airflow, Dagster, and Prefect. Strong Python and advanced SQL skills; proven track record managing data observability, column-level lineage, and data governance at scale. Experienced across AWS and GCP, with Infrastructure as Code via Terraform. Demonstrated use of AI/LLM tooling (Cursor, Claude) to accelerate engineering workflows and act as quality gate on AI-generated pipeline code. Effective remote-first collaborator and mentor with experience leading cross-functional Data & AI teams.

Core Competencies

Data Pipeline & ELT/ETL: Snowflake · Databricks · BigQuery · Redshift · dbt Core/Cloud · Fivetran · Airbyte · DLTHUB

Orchestration: Apache Airflow · Dagster · Prefect

Programming Languages: Python · PySpark · Scala · Java · SQL

Cloud Platforms: AWS (S3, EC2, Lambda) · GCP (GCS, GKE)

Infrastructure as Code: Terraform · Ansible · Docker · Kubernetes

Observability & Governance: Metaplane · Datadog · Column-level lineage · Data quality assurance

Data Modelling: Kimball dimensional modelling · OLAP/OLTP · Data lakes · Schema design

BI & Analytics: Sigma Computing · Looker · PowerBI

AI Tooling: Cursor · Claude · MLOps/DataOps practices

Leadership: Team mentoring · Agile/Scrum · Stakeholder management · Technical debt management

Professional Experience

Senior Data Engineer (Contract) | **Brainforge, USA** | *Dec 2025 – Present*

Magicspoon

- Built a GraphQL-based ELT pipeline ingesting 20M+ SPINS records (~30 GB) monthly into Amazon Redshift, orchestrated via Prefect Cloud — eliminating a 40-hour/month manual Excel export process and enabling Magicspoon to analyse brand performance across geographies, store networks, and competitors at weekly, monthly, and quarterly granularity.

CTA (Consumer Technology Association)

- Greenfield-built CTA's Snowflake data warehouse from scratch — designed RBAC model, provisioned warehouse objects, and ingested data from Salesforce Marketing Cloud (S3) and Shopify into a dbt-modelled dimensional layer, giving CTA their first unified, analytics-ready data platform for global events including CES.
- Engineered an identity stitching solution using rule-based logic and fuzzy string matching to resolve 100K+ event registrations (exhibitors) across inconsistently named entities (e.g. "Samsung Inc" vs "Samsung USA") to hundreds of CTA member companies — a problem that was previously unresolved and handled manually in Excel. Unlocked for the first time the ability to analyse member company attendance, per-company event spend, and attendance by geography.

Eden Health

- Configured Metaplane monitoring across Eden Health's BigQuery datasets — covering data freshness, dbt pipeline health, and anomaly detection (row counts, column-level aggregates) — shifting data issue detection from reactive (stakeholders discovering problems in dashboards) to proactive, eliminating silent pipeline failures going unnoticed.

- Built an end-to-end influencer performance pipeline ingesting data from Upfluence via Segment into BigQuery, modelled with dbt, and reverse-ETL'd into Northbeam via Dagster orchestration — replacing a manual Google Sheets process for tracking commissions and performance across ~50 active influencers and 1,000s of monthly orders. Enabled Eden Health to identify high-performing influencers by revenue contribution, directly informing contract renewal decisions and improving overall marketing ROI.
- Implemented Kimball-style dbt dimensional modelling across Snowflake, BigQuery, and Redshift for multiple clients; built self-service dashboards on Sigma Computing aligned to business KPIs.

Senior Data Engineer (Contract) | Haven Holidays – Bourne Leisure, UK | May 2025 – Dec 2025

- Diagnosed and resolved critical data integrity issues in Haven's Coupa procurement pipeline and dbt models that were producing irreconcilable results against the legacy ERPX system — covering millions of Purchase Orders and ~£40M in annual procurement spend. Eliminated manual Excel-based reconciliation and restored Finance team confidence in the data, unblocking procurement analytics that had been stalled due to untrusted figures.
- Reverse-engineered undocumented VAT calculation logic from Haven's legacy SQL Server system (original engineer having left the organisation) and recreated it entirely using transactional Snowflake data and dbt — achieving exact reconciliation with the legacy tax reports. Enabled Haven to retire dependence on the legacy system for VAT reporting and validated Snowflake as the authoritative source for financial compliance reporting.
- Enforced best practices for data modelling, pipeline reliability, and documentation in a remote async environment.

Senior Data Architect | BSI Financial (formerly Entra Solutions Pvt Ltd) | Jun 2023 – Dec 2025

- Led end-to-end modernisation of BSI Financial's data platform from a legacy SQL Server warehouse (T-SQL/PL-SQL procedures with no structured modelling) to a modern Databricks + Snowflake architecture — consolidating 8 heterogeneous sources (COBOL/SFTP exports, SQL Server, Snowflake, MongoDB, Postgres, Jira, Google Sheets, S3) into a unified dbt-modelled dimensional layer serving Finance, Collections, Call Centre, and Customer Service domains.
- Replaced three separate vendor solutions with in-house engineering — a \$40K/year report built on COBOL-sourced data, a \$70K/year Excel-based transformation tool (CAS) feeding LoanServ, and a \$25K/year CloverDX UI transformation solution — delivering \$135K in combined annual cost savings.
- Ingested and modelled customer chat data to build reporting on chatbot performance — revealing that the chatbot was resolving 75% of inbound queries, providing the business case to reduce customer service headcount by 50%.
- Modelled call log data from the Five9 Call Centre to identify peak and off-peak demand patterns, enabling the call centre manager to optimise agent scheduling — reducing customer wait times and improving overall service performance.
- Built a collection agent performance scoring model enabling BSI Financial to objectively reward high performers and eliminate underperforming agents — driving sustained improvement in collections performance.
- Designed the schema and data infrastructure underpinning BSI Financial's RAG-based chatbot solution, enabling AI-powered customer service at scale.
- Implemented full data observability and column-level lineage across all pipelines using Datadog; administered roles and permissions on Databricks and Snowflake via Terraform (Infrastructure as Code).
- Developed a DSL enabling transformations authored in Excel, compiled to PySpark — bridging business and engineering teams.

Data Reliability Engineering Manager | Shopify | Apr 2022 – Jun 2023

- Identified and eliminated excessive GCS storage costs through systematic bucket-level cost analysis using Google Studio — archiving and deleting stale logs and over-retained data to deliver \$80K in annual storage savings.
- Built end-to-end data lineage across all pipeline models, tagging each dataset with an owner — replacing fully manual downstream impact tracing with a single script execution. When pipelines failed, engineers could instantly identify affected downstream models and notify the right dataset owners, dramatically reducing time-to-communicate during incidents.

- Served as a core member of the Data Reliability team, performing daily root cause analysis on pipeline and model failures across a Google Kubernetes Engine (GKE) + Apache Airflow platform. Authored runbooks for recurring cluster-level failures, converting one-off investigative efforts into repeatable resolution playbooks.
- Managed technical debt to ensure scalability and system stability of Trino-based analytical platform; implemented Infrastructure as Code best practices using Terraform across data services.

Engineering Manager – Connectors | **Fivetran** | *Oct 2018 – Mar 2022*

- Led 3 teams of 15 engineers building and maintaining 20+ production ELT connectors across eCommerce, CRM, and HR domains (including Salesforce, Shopify, Stripe, NetSuite, SAP Concur, Microsoft Dynamics 365, and Xero) — maintaining 99.9% connector uptime across the portfolio.
- Identified that real-time streaming inserts into BigQuery were driving Fivetran's internal log ingestion costs to \$1,000/day — redesigned the pipeline to use batched ingestion, reducing costs to ~\$200/day and delivering an 80x cost reduction saving approximately \$292K annually.
- Delivered a customer-facing logs feature providing full pipeline transparency — giving customers real-time visibility into extraction start/end times, API request details, row counts, and data volumes at their chosen destination. Enabled customers to build automated actions and workflows triggered directly from log events.
- Resolved a chronic data integrity issue in the NetSuite Suite Analytics connector — one of Fivetran's highest-value connectors — by implementing a checksum-based verification mechanism between source and target warehouse. Eliminated a high volume of customer support tickets and restored customer confidence in connector reliability.
- Championed agile methodologies, improving developer productivity; decreased system errors by 40% through improved engineering practices; introduced backward sync, reducing data availability lag by weeks.

Principal Member of Technical Staff | **Oracle** | *Jun 2013 – Oct 2018*

- Designed and optimised REST Adapters for Oracle SOA Suite, enhancing authentication protocols and throughput performance.
- Led performance optimisation for JMS Adapters and managed the Enterprise Delivery Network (EDN).
- Transitioned build pipeline from ANT to Maven, significantly improving build efficiency and reproducibility.

Tech Lead | **Thomson Reuters Pvt Ltd** | *Jun 2009 – May 2013*

- Built and maintained the Novus Boolean Search indexing solution processing petabytes of legal data.
- Developed the Rotary index for wildcard search, substantially enhancing search capability and performance.
- Mentored a team of 8 engineers, fostering technical growth and delivery excellence.

Tech Lead | **Wipro Technologies** | *Jul 2005 – Jan 2009*

- Designed and implemented web-based solutions for Novartis Pharmaceuticals including Course Tracking and People Search applications.
- Developed IMAGE, an expenditure-tracking solution for Novartis associate healthcare professional interactions.

Education

Master of Technology (M.Tech) – Computer Science | BITS Pilani | 2015

Bachelor of Technology (B.Tech) – Computer Science | Sikkim Manipal University | 2003

Key Achievements

- Delivered \$135K in annual vendor cost savings at BSI Financial by replacing three separate paid solutions with in-house engineering.
- Reduced BigQuery log ingestion costs by 80x (~\$292K/year) at Fivetran through architectural pipeline redesign from streaming to batched inserts.
- Cut GCS storage costs by \$80K/year at Shopify via systematic data lifecycle optimisation.
- Eliminated 40 hours/month of manual Excel-based data work for Magicspoon by automating SPINS data ingestion at 20M+ records/month.
- Resolved previously unresolved identity stitching problem for CTA, mapping 100K+ event registrations to member companies for the first time.

- Achieved exact reconciliation of £40M procurement spend and complex VAT calculations at Haven Holidays — restoring Finance team trust in the data platform.
- Enabled 50% reduction in customer service headcount at BSI Financial by providing analytics proving 75% chatbot query resolution rate.
- Maintained 99.9% connector uptime across 20+ production ELT connectors at Fivetran, leading 3 teams of 15 engineers.